

EXPERIMENT (11)

AIM

To create and use AUTO_INCREMENT on the Hostel Management System tables.

DESCRIPTION

AUTO_INCREMENT automatically generates unique values for a primary key column.

SYNTAX

```
CREATE TABLE table_name(  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    column_name datatype  
);
```

Questions and Answers

1. Populate StudentID using AUTO_INCREMENT in Hostel_Student_Auto table.

```
DROP TABLE IF EXISTS Hostel_Student_Auto;
```

```
CREATE TABLE Hostel_Student_Auto(  
    StudentID INT AUTO_INCREMENT PRIMARY KEY,  
    StudentName VARCHAR(30),  
    Gender CHAR(1),  
    DOB DATE,  
    MobileNo BIGINT,  
    RoomID INT  
);
```

```
INSERT INTO Hostel_Student_Auto(StudentName,Gender,DOB,MobileNo,RoomID)  
VALUES  
(('Lokesh','M','2005-03-15',9876543210,101),  
(('Rahul','M','2004-07-10',9876543211,102),  
(('Priya','F','2005-01-25',9876543212,103);
```

```
SELECT * FROM Hostel_Student_Auto;
```

```
SQL File 4*  SQL File 5*  SQL File 6*  SQL File 7*  SQL File 8*  SQL File 7* x
Limit to 1000 rows
1 • USE HostelDB;
2
3 • DROP TABLE IF EXISTS Hostel_Student_Auto;
4
5 • CREATE TABLE Hostel_Student_Auto (
6     StudentID INT AUTO_INCREMENT PRIMARY KEY,
7     StudentName VARCHAR(30),
8     Gender CHAR(1),
9     DOB DATE,
10    MobileNo BIGINT,
11    RoomID INT
12 );
13
14
```

```
SQL File 4*  SQL File 5*  SQL File 6*  SQL File 7*  SQL File 8*  SQL File 7* x
Limit to 1000 rows
13
14 • If Execute the selected portion of the script or everything, if there is no selection
15 (StudentName, Gender, DOB, MobileNo, RoomID)
16 VALUES
17 ('Lokesh','M','2005-03-15',9876543210,101),
18 ('Rahul','M','2004-07-10',9876543211,102),
19 ('Priya','F','2005-01-25',9876543212,103);
20
21 • SELECT * FROM Hostel_Student_Auto;
```

Result Grid

	StudentID	StudentName	Gender	DOB	MobileNo	RoomID
▶	1	Lokesh	M	2005-03-15	9876543210	101
	2	Rahul	M	2004-07-10	9876543211	102
	3	Priya	F	2005-01-25	9876543212	103
*	NULL	NULL	NULL	NULL	NULL	NULL

Hostel_Student_Auto 2 x Apply Revert

2. Manually populate StudentID.

```
INSERT INTO Hostel_Student_Auto
(StudentID,StudentName,Gender,DOB,MobileNo,RoomID)
VALUES
(10,'Arun','M','2005-04-20',9876543213,101);
SELECT * FROM Hostel_Student_Auto;
```

SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 7* x

Limit to 1000 rows

```

19 ('Priya','F','2005-01-25',9876543212,103);
20
21 • SELECT * FROM Hostel_Student_Auto;
22 • INSERT INTO Hostel_Student_Auto
23 (StudentID, StudentName, Gender, DOB, MobileNo, RoomID)
24 VALUES
25 (10,'Arun','M','2005-04-20',9876543213,101);
26
27 • SELECT * FROM Hostel_Student_Auto;

```

Execute the selected portion of the script or everything, if there is no selection

Result Grid

	StudentID	StudentName	Gender	DOB	MobileNo	RoomID
▶	1	Lokesh	M	2005-03-15	9876543210	101
	2	Rahul	M	2004-07-10	9876543211	102
	3	Priya	F	2005-01-25	9876543212	103
	10	Arun	M	2005-04-20	9876543213	101
*	NULL	NULL	NULL	NULL	NULL	NULL

Hostel_Student_Auto 3 Hostel_Student_Auto 4 x Apply Revert

3. Drop the AUTO_INCREMENT property.

ALTER TABLE Hostel_Student_Auto
MODIFY StudentID INT NOT NULL;

SELECT * FROM Hostel_Student_Auto;

SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 7* x

Limit to 1000 rows

```

23 (StudentID, StudentName, Gender, DOB, MobileNo, RoomID)
24 VALUES
25 (10,'Arun','M','2005-04-20',9876543213,101);
26
27 • SELECT * FROM Hostel_Student_Auto;
28 • ALTER TABLE Hostel_Student_Auto
29 MODIFY StudentID INT NOT NULL;
30
31 • SELECT * FROM Hostel_Student_Auto;

```

Result Grid

	StudentID	StudentName	Gender	DOB	MobileNo	RoomID
▶	1	Lokesh	M	2005-03-15	9876543210	101
	2	Rahul	M	2004-07-10	9876543211	102
	3	Priya	F	2005-01-25	9876543212	103
	10	Arun	M	2005-04-20	9876543213	101
*	NULL	NULL	NULL	NULL	NULL	NULL

Hostel_Student_Auto 5 Hostel_Student_Auto 6 Hostel_Student_Auto 7 x Apply Revert

RESULT

Thus, AUTO_INCREMENT was successfully created, used and removed on the Hostel Management System table.